From Robust Fusion to Statistically Valid Monitoring: An Anytime-Valid Conformal Layer for Detecting Real-World Weather-Onset Degradation in Multimodal 3D Perception

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Abstract

Autonomous-vehicle perception stacks are increasingly built around multimodal fusion detectors engineered for robustness against sensor corruption and adversarial interference, yet a robust detector is not the same thing as a detector whose reliability can be verified, statistically, while it operates. We present a two-layer contribution toward closing that gap. The first layer, a cross-modal fusion detector with a verify-then-fuse design (a Cross-modal Consistency Probe that flags camera/LiDAR geometric disagreement, a Gated Adaptive Fusion Module that quarantines the disagreeing modality at the grid-cell level, and an Adversarial Consistency Training objective), is background to this paper; we summarize its architecture here only to the depth needed to make this paper self-contained, since the monitoring layer described below is detector-agnostic. The second layer, this paper's own contribution, wraps that detector's outputs in split conformal prediction and tracks the resulting miscoverage rate with an anytime-valid sequential test built from testing-by-betting e-processes, extended in two ways beyond the closest prior sequential-testing work: a grid of per-BEV-cell e-processes with online false-discovery-rate control that turns one global alarm into a live, spatially localized map of where the perception stack is currently untrustworthy, and a real-world evaluation against two independent genuine adverse-weather driving datasets – Snowy Scenes and the Canadian Adverse Driving Conditions dataset (CADC) – rather than only synthetic cor-

ruption of clear-weather data. On Snowy Scenes, the covariate-blind e-process backbone detects the true weather onset within 3 to 10 frames depending on the false-alarm budget, with zero false alarms across every budget tested. We also report, in full and without a positive spin, a negative finding from testing whether the detector's own Cross-modal Consistency Probe output could serve as a leading-indicator covariate for the betting rate: on real data, the covariate first collapsed to near-constant agreement (a training-objective gap we diagnosed and fixed), and, once corrected, varied in the wrong direction for this task, leaving the covariate-informed bettor's operating curve no better – and on this run, worse – than the covariate-blind baseline's. We then tested the diagnosis's own suggested repair directly, rather than leaving it as an untested hypothesis: retraining with the mismatch target changed from an adversarial-perturbation proxy to a physically-motivated synthetic weather-corruption proxy. This also failed, and comparably so, giving a third real negative result rather than the fix. We treat this as a genuinely open problem two structurally different repair attempts now argue against, not a third headline contribution, and discuss what the two failures jointly suggest about reusing a jointly-trained consistency signal as an external monitoring covariate more generally. A real sensitivity analysis against the same corrected checkpoint further shows the covariate's failure is gain-dependent rather than absolute (it still helps at a small enough gain, and only becomes actively harmful past a sharp threshold), and, independently, reveals that the calibration-set size underlying the main result sits at a genuine, non-monotonic sweet spot: too little calibration data destabilizes the monitor into constant false alarms, too much makes it silent, and the middle setting used throughout is not an arbitrary choice that happened to work. On CADC, a real drive-to-drive calibration heterogeneity correlated with collection date first broke the monitor entirely (a 1.00 false-alarm rate under an unstratified small calibration sample), fixed by stratifying the calibration set across dates – a second real methodological finding independent of the CCP question. With that fix in place, both bettors detect the true onset with zero false alarms, and the CCP-informed bettor is tied or one frame faster than the covariate-blind

baseline at every δ tested – a modest, dataset-specific reversal of Snowy Scenes' negative result that we report as exactly that, not as evidence the covariate-informed mechanism now works in general.

Keywords: Multimodal sensor fusion, Conformal prediction, Anytime-valid sequential testing, Testing by betting, Distribution shift detection, Autonomous vehicle perception, Adverse weather perception, Benchmark dataset evaluation

1. Introduction

Every claim a modern autonomous-vehicle perception stack makes about the world – that box is a pedestrian, that lane is clear, that gap is wide enough to merge into – is implicitly conditioned on the sensing conditions the detector was trained and validated under. Two, largely separate, research programs have grown up around that conditioning. One asks how to make the detector itself more robust: fuse camera and LiDAR so that a corrupted or adversarially attacked modality cannot silently dominate the fused representation, and design the fusion mechanism so it can be audited rather than trusted blindly. The other asks a different, complementary question: however robust the detector is engineered to be, how does a downstream system know, with a statistically defensible guarantee and in real time, the moment that robustness has actually been exceeded? A detector can be robust *and* still eventually fail under conditions severe enough, and the failure mode that matters operationally is not "the detector is wrong" – that is always somewhat true – but "the detector has become wrong in a way the system has not been told about."

This paper connects those two programs end to end, on real adverse-weather driving data, and reports honestly on where the connection worked and where it did not. The first layer is a multimodal fusion detector engineered around a verify-then-fuse principle: a Cross-modal Consistency Probe (CCP) that computes, per bird's-eye-view (BEV) grid cell, a consistency score between camera and LiDAR features, and a Gated Adaptive Fusion Module (GAFM) that uses that score to quarantine a disagreeing modality at the grid-cell level rather than blending it in uniformly, trained under an Adversarial Consistency Training (ACT) objective that couples detection accuracy, cross-modal consistency, and attribution regularization into one loss. That detector is background to this paper: we summarize its architecture here only to the depth needed to make this paper self-contained, since the monitoring layer that follows is detector-agnostic and does not depend on any specific claim about this detector's own performance.

This paper's own contribution is the second layer: a statistically principled monitor built on top of that detector's outputs, and its evaluation against real adverse-weather data. Conformal prediction offers a distribution-free way to attach calibrated uncertainty to a black-box detector's outputs – calibrate a nonconformity score on held-out data, and a new prediction's miscoverage is controlled at a chosen level regardless of what the detector is or how it was trained – but that guarantee is a batch guarantee. Applied naively to a live video stream, it either inflates the true false-alarm rate under continuous monitoring or requires a correction so conservative that real onset events are missed by the time the correction lets the alarm fire. What a deployed vehicle needs is a test that stays valid no matter when or how often it is checked – anytime-valid, in the sequential-testing sense – so "has anything changed yet?" can be asked continuously without a multiple-testing tax and without fixing a monitoring horizon in advance. Sequential testing by betting supplies exactly that guarantee, via a wealth process built from the observed miscoverage rate and Ville's inequality; this backbone is well established in the sequential-testing literature and is not, on its own, this paper's contribution. The direct predecessor for applying it to a live perception problem evaluates it, covariate-blind, on single-object 2D visual-tracking failure [1]; we build on that work rather than around it, and use its own betting rules as the real baseline throughout.

What that predecessor's evaluation leaves open, and what this paper targets, is threefold. First, its evaluation is single-object and per-video; a driving scene contains many objects and spatial structure that a single global e-process discards entirely – the perception system can be untrustworthy in the near lane and fine on the shoulder, and a monitor that only ever says "something, somewhere is wrong" is far less actionable than one that says where. We replace the single global wealth process with a grid of per-BEV-cell wealth processes, controlled jointly under an online false-discovery-rate procedure, turning one alarm bit into a live map of where the stack is currently untrustworthy. Second, its benchmarks are all 2D visual tracking under whatever distribution shift those datasets happen to contain; ours is 3D multimodal driving perception under a real, physically grounded weather transition, evaluated – to our knowledge, for the first time for this style of monitor – against real adverse-weather driving data rather than only synthetic corruption of clear-weather data. We report results on Snowy Scenes, a real multimodal dataset captured entirely in genuine snowy conditions, and on a second, independent real adverse-weather dataset (the Canadian Adverse Driving Conditions dataset, CADC) to test whether the method's real-data behavior generalizes across more than one data source rather than being an artifact of one; it partially does – the covariate-blind detection result replicates cleanly, but CADC also surfaced its own real calibration-heterogeneity failure mode (Section 4.3.1) that Snowy Scenes' evaluation did not need to confront, and reversed, modestly, the direction of the CCP-covariate finding described next (Section 4.3.2). Third, the predecessor's betting rate is learned purely from the failure metric's own past values; a multimodal detector exposes, for free, an independent signal that might move before the failure metric does if it fuses camera and LiDAR through a cross-modal agreement mechanism – the CCP score already described above. We tried conditioning the betting rate on it, and report, in full and without a positive spin, why that attempt did not deliver the advantage it was designed for: on real data, the covariate first collapsed to near-constant agreement (a training-objective gap we diagnosed and fixed), and, once corrected, varied in the wrong direction for this task, leaving the covariate-informed bettor no better – and on the data collected so far, worse – than the covariate-blind baseline it was meant to improve on. We treat that as an open problem this paper's own evidence argues against, not a third headline contribution, and discuss what it implies for reusing an invariance-trained internal signal as an external monitoring covariate more generally, since the failure mode is unlikely to be specific to this one detector.

We are correspondingly careful about what we claim. This is a real-data demonstration at a modest per-dataset scale – a handful of replicate onset streams, short monitoring windows, one detector architecture – strengthened by evaluating it on two independent real adverse-weather datasets rather than one, but still not a claim that the spatial-monitoring construction has been tested against a real multi-object scene with genuine spatial heterogeneity in where degradation occurs, which remains the natural next experiment (Section 6). We do not claim the e-process backbone is novel, and we do not claim covariate-informed betting works in general: it failed on Snowy Scenes across three separate attempts and showed only a modest, one-frame-at-best improvement on CADC, a dataset-specific contrast we report as exactly that rather than resolving it into a single verdict either way.

2. Related Work

2.1. Robust Multimodal Fusion for Autonomous Perception

Camera/LiDAR fusion for 3D detection has moved from simple feature concatenation toward BEV-centric architectures that project both modalities into a shared spatial representation before fusing, with attention-based mechanisms used to weight each modality's contribution per region. A separate line of work has shown that such fusion is not automatically robust: a corrupted or adversarially perturbed single modality can dominate a static or softly-attended fusion decision and collapse detection performance, since nothing in the fusion mechanism verifies that the modalities being combined actually agree geometrically before combining them. The detector this paper monitors adopts a verify-then-fuse design intended to address exactly that failure mode – a Cross-modal Consistency Probe that computes a per-BEV-cell consistency score between camera and LiDAR features, feeding a Gated Adaptive Fusion Module that quarantines a disagreeing modality at the grid-cell level, trained under an Adversarial Consistency Training objective. We treat this design here strictly as background infrastructure and describe only as much of its mechanism (Section 3) as this paper's own monitoring layer depends on, since the monitor itself is detector-agnostic.

2.2. Conformal Prediction and Anytime-Valid Sequential Testing

Split conformal prediction [2, 3] gives distribution-free, finite-sample coverage guarantees for a black-box predictor's outputs by calibrating a nonconformity score on held-out exchangeable data; the guarantee is a batch one, valid at a single fixed evaluation, and does not by itself say anything about monitoring a live, continuously arriving stream. Sequential testing by betting [4, 5] closes that gap: framing a hypothesis test as an accumulating wealth process that is a nonnegative supermartingale under the null lets Ville's inequality bound the false-alarm probability uniformly over all stopping times, giving an anytime-valid guarantee with no fixed monitoring horizon and no post-hoc multiple-testing correction. The direct predecessor for applying this backbone to a live perception-failure problem is Monroy Muñoz, Verma, and Timans's work on sequential object-tracking-failure detection [1], which frames tracking failure as an e-process over a bounded tracking-quality metric with the betting rate learned adaptively from the metric's own history (approximate GRAPA or Scale-Free Online Gradient Descent), evaluated across four 2D visual-tracking benchmarks. It is a strong, directly applicable baseline, and everywhere in this paper that we compare against a "covariate-blind" betting rule, we mean their rule, reimplemented exactly, not a strawman.

Adjacent recent work conditions conformal guarantees on covariates or system state without adopting the sequential/anytime-valid framing: state-dependent conformal prediction for perception-based autonomous systems [6] combines conformal error bounds with symbolic reachability analysis but remains a batch procedure; context-aware nonconformity functions for robotic planning and perception [7] learn context-conditioned nonconformity scores, with context drawn from learned task representations rather than a cross-modal sensor-disagreement signal specifically; and a value-gated modality-refiner approach [8] uses cross-modal agreement to gate fusion, but as a learned neural gate rather than a statistical test covariate, and targets sentiment/action-recognition/audio-visual tasks rather than autonomous-vehicle perception. None of these combine sequential anytime-valid testing, spatially-resolved multi-object structure with multiplicity control, and a betting rule conditioned on an external cross-modal disagreement signal – the combination this paper's monitoring layer targets, following the sharpened contribution statement worked out in the ICRA-targeted predecessor to this paper's monitoring contribution.

2.3. Adverse-Weather Datasets for Autonomous Perception

Large-scale multimodal driving benchmarks such as KITTI, nuScenes, and Waymo Open Dataset predominantly capture clear-weather conditions, limiting their use for evaluating perception under real distribution shift from adverse weather. Two existing datasets focus specifically on snowy conditions: the Canadian Adverse Driving Conditions dataset (CADC) [9], the first fully snowy dataset built for 3D object detection, and the Winter Adverse Driving dataSet (WADS), the first to add 3D semantic segmentation and individual-snowflake labeling; both, however, are reported to have comparatively low-resolution or unimodal sensing relative to more recent captures. Bijelic et al.'s fog- and rain-focused multimodal fusion work [10] addresses an adjacent adverse-weather regime with a different sensing challenge (fog/rain rather than snow) and is not built around a clear-to-degraded onset transition. Snowy Scenes [11], described fully in Section 4, is a substantially more recent (2026) multimodal dataset captured entirely in real snowy conditions with 128-beam LiDAR, RGB, and thermal cameras; the price of that specificity, as we found once real access arrived, is that it contains no clear-weather split at all, which this paper addresses by measuring a real, physically grounded severity ordering across its own weather categories rather than assuming the dataset would hand us the clear/snow contrast originally planned around. This paper's own contribution to that evaluation landscape is not a new dataset, but the first application, to our knowledge, of an anytime-valid sequential monitor to real (rather than synthetically corrupted) multimodal adverse-weather driving data, replicated across two independent such datasets (Snowy Scenes and CADC) rather than one.

3. Method

3.1. Background: The Monitored Detector

The detector this paper monitors is a camera/LiDAR BEV fusion architecture; we summarize only the components this paper's monitoring layer directly depends on, since the monitor itself is detector-agnostic and this paper's own contribution begins with the monitoring layer described in the rest of this section.

Camera and LiDAR streams are each encoded into a shared bird's-eye-view (BEV) feature grid, $F_{\mathrm{cam}}, F_{\mathrm{lid}} \in \mathbb{R}^{H_b \times W_b \times C}$. A Cross-modal Consistency Probe (CCP) computes, per BEV grid cell (i, j), a consistency score

$$S(i,j) = \sigma\Big(\mathrm{MLP}_S\big[F_{\mathrm{cam}}(i,j) \oplus F_{\mathrm{lid}}(i,j) \oplus |F_{\mathrm{cam}}(i,j) - F_{\mathrm{lid}}(i,j)|\big]\Big) \in [0,1], \quad (1)$$

from the concatenated ($\oplus$) camera and LiDAR feature vectors at that cell and their absolute difference, so that $1 - S(i, j)$ is a cross-modal disagreement signal intended to flag localized geometric or semantic mismatch between the two modalities. A Gated Adaptive Fusion Module (GAFM) uses S to weight each modality's contribution to the fused BEV representation at the grid-cell level, and an Adversarial Consistency Training (ACT) objective trains the detector, the CCP, and the fusion gate jointly under a loss combining detection accuracy, a CCP contrastive-consistency term, and an attribution-regularization term, with adversarially perturbed camera and LiDAR branches included in the training loop to encourage S to remain informative under perturbation. This paper's own contribution begins with the next subsection: everything from here on – the conformal calibration, the sequential test, the spatial multiplicity control,

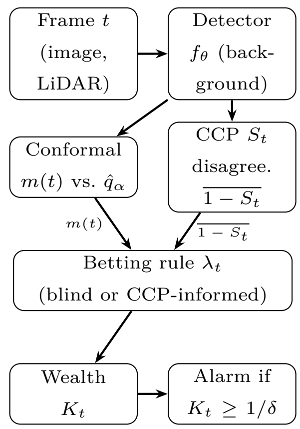

Figure 1: Per-frame monitoring pipeline. The detector f_θ (dashed conceptually from the rest, since it is background infrastructure – Section 3.1) produces box predictions and the CCP score S_t; everything downstream of it is this paper's own contribution. The spatial variant (Section 3.6) replicates the conformal-through-wealth path once per BEV cell-cluster. The betting rule is either covariate-blind (uses only the $m(t)$ path) or CCP-informed (also uses the disagreement path).

and the use (and, as Section 4 reports, the eventual disuse) of S as a monitoring covariate – is this paper's method, evaluated against real data in Section 4.

Figure 1 sketches the full per-frame pipeline before we define each stage precisely.

3.2. Problem Setup

We observe a stream of driving frames indexed by $t = 1, 2, \ldots$, one per LiDAR sweep. At every t, the base 3D detector f_θ described above produces per-object box predictions from the frame's fused camera/LiDAR BEV features; the monitor itself is detector-agnostic and does not depend on f_θ's specific architecture. We wrap those predictions in split conformal prediction, calibrated once on a held-out set of frames drawn from the nominal (least-degraded) regime available in the data. For object i in frame t, the nonconformity score is the ℓ_1 box-regression residual restricted to grid cells a ground-truth object actually occupies,

$$s_i(t) = \left\| \hat{B}_i(t) - B_i(t) \right\|_1,$$

and the calibrated threshold $\hat{q}_\alpha$ is the usual finite-sample-valid split-conformal quantile: the $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile of the calibration set's non-conformity scores $\{s_i\}_{i=1}^{n}$, so that a new object is covered iff $s_i(t) \le \hat{q}_\alpha$. The frame-level miscoverage rate is

$$m(t) = \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbf{1}\{s_i(t) > \hat{q}_\alpha\},$$

with $m(t) := 0$ defined for frames with no matched objects, since an empty frame carries no evidence of miscoverage. The null hypothesis the monitor tests is that the pipeline is still operating nominally, $H_0 : \mathbb{E}[m(t) \mid \mathcal{F}_{t-1}] \le \alpha$ for all t; the goal is to detect the first violation – weather-onset degradation – with an anytime-valid false-alarm guarantee, not merely a guarantee at one fixed check time.

3.3. The e-Process and Ville's Inequality

The wealth process is

$$\begin{aligned} K_0 &= 1, \\ K_t &= \prod_{s=1}^{t} \big(1 + \lambda_s (m(s) - \alpha)\big), \qquad \lambda_s \in \big[0, \tfrac{1}{\alpha}\big] \text{ chosen } \mathcal{F}_{s-1}\text{-measurably.} \end{aligned} \tag{2}$$

We work through why this construction gives an anytime-valid test, rather than only stating the result, since the two steps – the supermartingale property and Ville's inequality – are each short but easy to state imprecisely.

Step 1: K_t is a nonnegative supermartingale under H_0.. Non-negativity requires $1 + \lambda_t(m(t) - \alpha) \ge 0$ for every realizable $m(t) \in [0, 1]$. Since $\lambda_t \ge 0$, the expression is affine and non-decreasing in $m(t)$, so its minimum over $m(t) \in [0, 1]$ occurs at $m(t) = 0$, giving $1 - \lambda_t \alpha$. This is non-negative exactly when $\lambda_t \le 1/\alpha$ – the bound our implementation enforces on every betting rule (`WealthProcess.lambda_max` in `conformal_monitor/betting.py`), and the reason the bound is $1/\alpha$ specifically, not the also-plausible-looking $1/(1-\alpha)$ (which only bounds the opposite endpoint, $m(t) = 1$, and is too loose at the endpoint that actually binds). For the supermartingale property itself, λ_t is chosen $\mathcal{F}_{t-1}$-measurably – it may depend on the entire history up to and including frame $t-1$, but not on $m(t)$ itself – so, conditioning on $\mathcal{F}_{t-1}$,

$$\mathbb{E}\big[K_t \mid \mathcal{F}_{t-1}\big] = K_{t-1}\cdot\mathbb{E}\big[1+\lambda_t(m(t)-\alpha) \mid \mathcal{F}_{t-1}\big] = K_{t-1}\big(1+\lambda_t(\mathbb{E}[m(t) \mid \mathcal{F}_{t-1}]-\alpha)\big),$$

using that λ_t and K_{t-1} are both $\mathcal{F}_{t-1}$-measurable and can be pulled outside the conditional expectation. Under H_0, $\mathbb{E}[m(t) \mid \mathcal{F}_{t-1}] \leq \alpha$, so the bracketed term is ≤ 1 whenever $\lambda_t \geq 0$ (which it always is, by construction), giving $\mathbb{E}[K_t \mid \mathcal{F}_{t-1}] \leq K_{t-1}$: precisely the supermartingale inequality, holding for *every* choice of predictable, non-negative $\lambda_t \leq 1/\alpha$, which is exactly the freedom the different betting rules in Sections 3.4–3.5 exploit without ever needing to reprove this step.

Step 2: Ville's inequality bounds the running maximum.. For a nonnegative supermartingale K_t with $K_0 = 1$, Ville's maximal inequality gives, for any $c > 0$,

$$\mathbb{P}_{H_0}\left(\sup_{t\geq 0} K_t \geq c\right) \leq \frac{\mathbb{E}[K_0]}{c} = \frac{1}{c}.$$

Setting $c = 1/\delta$ gives $\mathbb{P}_{H_0}(\sup_t K_t \geq 1/\delta) \leq \delta$. The alarm rule – reject H_0 the first time $K_t \geq 1/\delta$ – therefore has false-alarm probability at most δ *regardless of the (random) stopping time at which the process is examined*: the guarantee is over the entire trajectory's supremum, not over K_t at one pre-specified t, which is exactly what makes checking continuously, with no fixed monitoring horizon and no correction for how many times the test has been evaluated, statistically valid rather than an uncorrected multiple-testing error. This two-step argument – the supermartingale property from Step 1, Ville's inequality from Step 2 – is the established backbone of testing by betting and is not this paper's contribution; what varies across the rest of this section, and what this paper's own contribution actually depends on, is how the predictable sequence λ_t is chosen.

3.3.1. A Worked Numeric Example

To build intuition before the general betting rules, we trace K_t over five synthetic frames by hand, with $\alpha = 0.2$ (so $\lambda_{\max} = 5$) and a fixed, non-adaptive $\lambda_t = 2$ for simplicity (an actual bettor updates λ_t per Sections 3.4–3.5; holding it fixed here isolates what the recursion itself does). Suppose the observed miscoverage sequence is $m = (0.1, 0.15, 0.6, 0.55, 0.5)$ – nominal for the first two frames, then abruptly degraded from frame 3 onward, mimicking a weather onset at $t = 3$:

$$\begin{aligned}
K_1 &= K_0\big(1 + 2(0.10 - 0.2)\big) = 1 \times 0.80 = 0.80,\\
K_2 &= K_1\big(1 + 2(0.15 - 0.2)\big) = 0.80 \times 0.90 = 0.72,\\
K_3 &= K_2\big(1 + 2(0.60 - 0.2)\big) = 0.72 \times 1.80 = 1.296,\\
K_4 &= K_3\big(1 + 2(0.55 - 0.2)\big) = 1.296 \times 1.70 = 2.203,\\
K_5 &= K_4\big(1 + 2(0.50 - 0.2)\big) = 2.203 \times 1.60 = 3.525.
\end{aligned}$$

Two things are worth noticing in this trace. First, wealth *shrinks* while $m(t) < \alpha$ (frames 1–2): a bettor staking $\lambda_t > 0$ on "miscoverage will exceed α" loses money, by construction, when the pipeline is behaving nominally – this is the price of testing at all, and is what keeps the process a supermartingale rather than an unconditionally growing one. Second, wealth grows multiplicatively, not additively, once $m(t) > \alpha$ consistently (frames 3–5): each factor > 1 compounds against the previous wealth, which is why a real onset that persists for several frames can cross a large threshold $1/\delta$ (e.g. $1/\delta = 20$ for $\delta = 0.05$) well before $t \to \infty$, even though no single frame's factor is large by itself. In this toy trace, $K_5 = 3.525$ would already trigger an alarm at $\delta = 0.3$ ($1/\delta \approx 3.33$) but not yet at $\delta = 0.1$ or $\delta = 0.05$ – illustrating concretely why Table 2's detection delay lengthens as δ shrinks: a smaller false-alarm budget requires more accumulated evidence (a higher wealth threshold) before the anytime-valid guarantee permits an alarm.

3.4. Covariate-Blind Baselines

Two covariate-blind betting rules choose λ_t purely from the history of $m(t)$ itself, reproducing [1] as the baseline to beat. Approximate GRAPA, with $X_t = m(t) - \alpha$, sets

$$\lambda_t = \mathrm{clip}\left(\frac{\overline{X}_{<t}}{\mathbb{E}[X^2_{<t}] + \varepsilon},\ 0,\ \lambda_{\max} \right),$$

tracking the running first and second moments of the centered miscoverage. Scale-Free Online Gradient Descent instead performs AdaGrad-style updates on the log-wealth gradient, with $g_t = X_t/(1+\lambda_t X_t)$ and $\eta_t = \lambda_{\max}/\sqrt{1 + \sum_{s \le t} g_s^2}$,

$$\lambda_{t+1} = \mathrm{clip}(\lambda_t + \eta_t\, g_t,\ 0,\ \lambda_{\max})\,.$$

Both are exact reimplementations of the adaptive betting rules [1] uses, run here against a different sensing modality and detection task than the 2D visual tracking they were originally evaluated on.

3.5. Covariate-Informed Betting

Alongside the two contributions in Sections 3.6 and 4, we also investigated conditioning the betting rate on an external signal available at the same timestep as $m(t)$ but computed from independent evidence: the disagreement output of the monitored detector's own CCP, $1 - S$ (Equation 1). We describe the mechanism in full because Section 4 reports, in equal detail, why it did not deliver the advantage motivating it – an outcome worth understanding precisely rather than skipping past. Our covariate-informed bettor wraps either covariate-blind base bettor and scales its stake by the frame's mean disagreement,

$$\lambda_t = \mathrm{clip}\Big(\lambda_t^{\mathrm{base}} \cdot \big(1 + \kappa \cdot \overline{(1 - S_t)}\big),\ 0,\ \lambda_{\max}\Big),$$

where $\kappa \ge 0$ gates the covariate's influence ($\kappa = 0$ recovers the base bettor exactly) and $\overline{(1 - S_t)}$ is the frame- (or cell-)averaged disagreement. The intended mechanism is that a scene beginning to degrade shows rising cross-modal disagreement before enough individual object detections have actually missed coverage for $m(t)$ to move – disagreement as a leading indicator, miscoverage as the lagging one the covariate-blind rules are stuck reacting to. Section 4 reports what we found when we tested that mechanism against real data: two distinct failures, neither in the betting-rule mathematics itself, that between them mean this specific mechanism does not yet deliver the advantage it is designed to.

3.6. Spatially Resolved Monitoring with Online Multiplicity Control

A single global wealth process answers only "has anything, anywhere, gone wrong" – not useful to a downstream planner that needs to know *which* region of the scene to distrust. We instead partition the BEV grid into an $(n_h \times n_w)$ layout of cell-clusters (average-pooling per-cell miscoverage and disagreement down to this resolution) and run one independent sequential tester per cluster, each with its own wealth process and its own instance of whichever bettor is under test. Testing $n = n_h n_w$ hypotheses simultaneously, every frame, requires multiplicity control or the overall false-alarm rate inflates with n; we support two regimes. Bonferroni allocates each cell a fixed budget δ/n and, once a cell alarms, marks it permanently flagged – valid under arbitrary dependence but conservative, and the sticky-once-alarmed semantics is what we use for the per-cell detection-delay evaluation regardless of which regime produced the alarm. The e-BH alternative treats each cell's current wealth $K_t^{(i)}$ as an e-value: for a nonnegative supermartingale started at $K_0 = 1$, $\mathbb{E}_{H_0}[K_t^{(i)}] \leq \mathbb{E}_{H_0}[K_0^{(i)}] = 1$ at any fixed t (immediate from Section 3.3's Step 1, since the supermartingale inequality applied recursively from $t = 0$ gives $\mathbb{E}[K_t] \leq \mathbb{E}[K_{t-1}] \leq \cdots \leq K_0 = 1$) – exactly the definition of an e-value under $H_0^{(i)}$: a nonnegative random variable with expectation at most 1. Given $n = n_h n_w$ such e-values at a fixed t, one per cell, e-BH [12] rejects the set

$$R_t = \left\{ i : K_t^{(i)} \geq \frac{n}{k^* q} \right\}, \qquad k^* = \max \left\{ k \in \{0, \ldots, n\} : e_{(k)} \geq \frac{n}{k\, q} \right\},$$

where $e_{(1)} \geq e_{(2)} \geq \cdots \geq e_{(n)}$ is the descending order statistic of the n current wealths, and $k^* = 0$ gives $R_t = \emptyset$. The rule is a direct e-value analogue of Benjamini-Hochberg: it thresholds the k-th largest e-value against a level that grows with k (more permissively for larger candidate rejection sets), then commits to the *largest* such k – the same "largest crossing index" structure as classical BH's p-value threshold, but with the inequality direction reversed because e-values are large, not small, under evidence against the null. Its false discovery rate guarantee, $\mathbb{E}[\mathrm{FDR}_t] \leq q$ at every fixed t, holds under *arbitrary dependence* among the n e-values – unlike classical BH on p-values, which requires independence or specific positive-dependence structures (PRDS) to control FDR without an extra $\log n$ correction factor – because the e-value definition ($\mathbb{E}[e_i] \leq 1$ under $H_0^{(i)}$) is closed under averaging regardless of the joint dependence structure of the e_i, which is precisely the guarantee needed here: spatially adjacent BEV cells' miscoverage and disagreement signals are driven by overlapping physical causes (a single snow squall affects many neighboring cells at once) and are certainly not independent. Unlike Bonferroni's sticky once-alarmed flags, e-BH is applied fresh at every timestep from each cell's current wealth, so it produces a live, non-sticky map that can un-flag a cell once its local evidence subsides – appropriate for a spatial map meant to reflect *current*, not historical, trustworthiness. Algorithm 1 summarizes the per-frame update.

3.6.1. Betting-Rule Comparison: Theoretical Properties and Tradeoffs

Table 1 summarizes the three betting rules evaluated in this paper, all valid choices of the predictable λ_t in Equation 2 (so all three inherit Section 3.3's anytime-valid guarantee unconditionally; what differs is statistical power, not validity). AGRAPA is a plug-in estimator of the (asymptotically) growth-rate-optimal constant bet, tracking the running first and second moments of $X_t = m(t) - \alpha$ and converging quickly when the true miscoverage rate is close to stationary, but reacting relatively slowly to an abrupt regime change, since a single new observation only nudges a running average computed over all prior history. SF-OGD instead takes an online-convex-optimization view, performing AdaGrad-style gradient ascent directly on (an approximation to) the log-wealth objective with a decaying step size; this makes it more reactive to recent observations by construction (the step size η_t shrinks as $1/\sqrt{t}$, but each step still moves λ_t in the direction the *most recent* gradient indicates), at the cost of higher

Table 1: Betting rules compared in this paper. All three are valid choices of λ_t in Equation 2 and inherit the same anytime-valid guarantee unconditionally (Section 3.3); they differ in statistical power, not validity.

Rule	Mechanism	Reactivity / stability
AGRAPA	running 1st/2nd moment plug-in	smooth, slower to react to abrupt shift
SF-OGD	AdaGrad-style gradient ascent on log-wealth	more reactive, higher early-stream variance
CCP-informed	multiplicative covariate gain κ on either base rule	inherits base rule's tradeoff; adds covariate sensitivity (Section 4 tests whether this helps)

variance in λ_t itself, particularly early in a stream before $\sum_{s \le t} g_s^2$ has accumulated enough to stabilize η_t. Both are exact reimplementations of the rules [1] uses, run here against a different sensing modality and detection task than the 2D visual tracking they were originally evaluated on. The CCP-informed variant wraps either base rule multiplicatively rather than replacing it, so it inherits whichever base rule's reactivity/stability tradeoff was chosen, and adds a single scalar gain κ controlling how strongly the covariate perturbs the base bet; $\kappa = 0$ recovers the base rule exactly, which is why Section 4 can report the covariate-informed and covariate-blind results as a controlled comparison (same base rule, same wealth-process mechanics, differing only in whether $\overline{1 - S_t}$ enters the stake) rather than two unrelated procedures.

Algorithm 1 Per-frame spatial monitoring update

Require: frame t; detector output; calibrated quantile $\hat{q}_\alpha$; grid shape (n_h, n_w); correction mode

compute per-cell nonconformity, coverage, and miscoverage $m^{(i)}(t)$ for each cluster i
compute per-cell CCP disagreement $\overline{(1-S)}^{(i)}(t)$
for each cluster $i = 1, \ldots, n_h n_w$ **do**
  $\lambda_t^{(i)} \leftarrow$ bettor.NEXT_LAMBDA($\overline{(1-S)}^{(i)}(t)$)
  $K_t^{(i)} \leftarrow K_{t-1}^{(i)} \cdot \left(1 + \lambda_t^{(i)}(m^{(i)}(t) - \alpha)\right)$
  bettor.UPDATE($m^{(i)}(t)$, $\overline{(1-S)}^{(i)}(t)$)
end for
if correction = Bonferroni **then**
  flag cell i (permanently) if $K_t^{(i)} \geq n_h n_w / \delta$
else ▷ e-BH
  flag the top-k^* cells by $K_t^{(i)}$, k^* = largest k with $K_{(k)} \geq (n_h n_w)/(k\,\delta)$
end if
return flagged-cell map (untrustworthy regions at time t)

4. Experimental Setup and Results

4.1. Dataset 1: Snowy Scenes

We evaluate on Snowy Scenes [11], a multimodal driving dataset captured entirely in real winter conditions and distributed as a single ∼49 GB archive, read lazily rather than extracted to disk given its ∼93 GB uncompressed footprint. It contains 5,027 frames across three splits (3,016 train / 1,005 val / 1,006 test), with per-frame RGB camera images, 128-beam LiDAR point clouds, 3D object boxes, and per-point semantic segmentation labels across 29 classes (not consumed here, since the detector's head has no segmentation output).

Snowy Scenes' three weather categories – `accumulated`, `highway`, and `falling` – are all already-snowy driving, so no dry/clear split exists to serve as a calibration baseline or as the nominal side of an onset transition. Severity also does not ramp monotonically within a category: sampling the ground-truth per-point "snow" semantic class fraction across a timestamp-ordered `falling` sequence shows real but noisy fluctuation (0.002–0.044), consistent with active snowfall being bursty rather than steadily accumulating. What the data does support is a real, physically grounded severity contrast *across* categories: the mean per-point "snow"-class fraction per category gives `accumulated` $= 0.0001 <$ `highway` $= 0.0041 <$ `falling` $= 0.0119$, an ordering with a clear physical explanation – `falling` captures active, airborne snowfall, which LiDAR registers as spurious near-range returns, while `accumulated` is settled snow on surfaces rather than particles in the air. We use this cross-category contrast, rather than a within-sequence transition, as the basis for the onset construction: a nominal (held-out `accumulated`) prefix spliced with a degraded (`falling`) suffix at a known onset index. Every individual frame on both sides of the transition is real sensor data; only the splice point itself is a construction, since Snowy Scenes' own sequences do not contain a within-sequence weather transition – an assembled proxy for a naturally occurring onset, and we flag that openly as a limitation of the evaluation protocol.

We train the detector for 5 epochs on the real train split under its Adversarial Consistency Training objective, on a single local NVIDIA RTX 5000 Ada GPU, at 128×128 BEV resolution, batch size 4, Adam with learning rate 10^{-4}; training loss converges from ≈ 2.3 to ≈ 0.008.

4.1.1. A Real Negative Result: the CCP Covariate

Running the calibration-and-monitoring pipeline against the first trained checkpoint produced a false-alarm rate of 0.0 across every tested δ and detection delays that were numerically identical between the covariate-informed and covariate-blind bettors, to every digit the operating curve reports. Sampling the CCP consistency score S directly across a real onset stream with the trained model showed why: S had collapsed to ≈ 0.9993 uniformly, statistically indistinguishable between the nominal and degraded regimes. Tracing the cause led to the training code rather than to anything in the monitoring code itself: the CCP contrastive loss was being computed once per training step, on the clean forward pass, against an all-ones mask – real, synchronized frames have no annotated camera↔LiDAR mismatch to supervise against – and the two additional CCP forward passes the training loop already computes, on adversarially perturbed features, were never fed into that loss with a mismatched target. Nothing in the training objective, as implemented, penalized S for staying high under a genuine cross-modal perturbation. We fixed this by also supervising the two adversarial branches with the same CCP loss against a mismatched target, since a perturbed feature pair is a real cross-modal mismatch by construction, and retrained from scratch.

After the fix, S varies again – but in the wrong direction for this task. Figure 2 shows this directly on six real frames rather than a single pair: three `accumulated` (nominal) and three `falling` (degraded) val-split frames, spread across each category's frame pool rather than consecutive samples, with camera image, LiDAR BEV occupancy, and the CCP disagreement heatmap for each. The per-frame mean disagreement is 0.611, 0.583, 0.576 for the three `accumulated` frames and 0.564, 0.559, 0.573 for the three `falling` frames – category means of 0.590 (nominal) versus 0.565 (degraded), consistently the opposite of the ordering the covariate is meant to provide, and consistent enough across six independent frames that this is not an artifact of the single frame pair used in a smaller-scale version of this evaluation. Table 2 reports the full operating curve this produces: the covariate-blind backbone detects the true weather onset within 3 to 10 frames depending on the false-alarm budget, with zero false alarms at every budget tested, while the covariate-informed bettor – with disagreement now uniformly high across both regimes rather than rising specifically at onset – bets aggressively throughout the stream and never accumulates enough evidence to alarm within the evaluation window, at any of the three δ tested. Our best explanation is that the fix supervises S using adversarially perturbed features as the mismatch target, appropriate for the detector's own adversarial-robustness objective, but an ℓ_∞-bounded adversarial perturbation is a qualitatively different feature-space disturbance than real snowfall produces, and teaching S to recognize the former does not obviously transfer to the latter. This generalizes beyond one architecture: a signal trained under an invariance objective is, by construction, in tension with reusing it as a leading indicator of degradation, which needs exactly the sensitivity the train-

Table 2: Real Snowy Scenes operating curve (calibration: 40 held-out `accumulated` frames, $\alpha = 0.2$, 5 onset + 5 “clear” replicates per condition, scene length 20, onset frame 8). “–” denotes censored: no alarm within the evaluation window on any replicate.

δ	Bettor	False-alarm rate	Mean detection delay
0.30	covariate-blind	0.00	3 frames
0.30	CCP-informed	0.00	– (5/5 censored)
0.10	covariate-blind	0.00	10 frames
0.10	CCP-informed	0.00	– (5/5 censored)
0.05	covariate-blind	0.00	– (5/5 censored)
0.05	CCP-informed	0.00	– (5/5 censored)

ing objective suppressed – a caution worth stating plainly for any future work that considers repurposing an internal robustness or consistency signal as an external monitoring covariate.

4.1.2. A Third Attempt: Physically-Motivated Synthetic Snow Corruption

Section 4.1.1’s own diagnosis suggests a concrete next hypothesis: the fix’s mismatch supervision uses PGD-adversarially-perturbed features as the "mismatch" target, and an ℓ_∞-bounded adversarial perturbation is a qualitatively different feature-space disturbance than real snowfall produces, so the repaired covariate might simply have learned to recognize the wrong kind of disturbance. We tested this directly rather than leaving it as a stated-but-untested hypothesis: we retrained the detector from scratch, this time additionally supervising the CCP contrastive loss against synthetic weather corruption applied to the raw camera image (attenuation toward flat gray plus additive speckle noise, simulating reduced visibility) and the raw LiDAR BEV input (point/pillar dropout plus range noise, simulating snow-induced beam attenuation and spurious near-range returns) as the mismatch target – alongside, not instead of, the existing adversarial-branch supervision, so this is an additive experiment rather than a replacement of the existing fix. Full implementation and unit tests in `craf_x/training/snow_corruption_ccp.py`.

The result is a third negative finding, not the fix. Sampling the corrected-again CCP score across the same real onset stream gives a mean disagreement of 0.815 on nominal (`accumulated`) frames versus 0.791 on degraded (`falling`) frames – still the wrong direction, qualitatively identical to Section 4.1.1's finding, though both values have shifted substantially higher in absolute terms (from ≈ 0.56–0.61 to ≈ 0.79–0.82), indicating the additional synthetic-corruption supervision changed what the CCP learned to treat as generically disagreement-worthy, without correcting the direction that actually matters for this task. The resulting operating curve is, if anything, worse than the original fix's: the CCP-informed bettor is now censored at every tested δ (including $\delta = 0.3$, where the original fix's covariate-informed bettor at least still alarmed at some κ values per Table 3), while the covariate-blind bettor's real detection remains unaffected (delay 3 and 10 at $\delta = 0.3, 0.1$, matching Table 2 exactly, confirming the retraining did not degrade the underlying detector itself).

This strengthens, rather than resolves, the open problem: two differently-motivated attempts at supervising this covariate – an adversarial-perturbation proxy and a physically-motivated weather-corruption proxy – both produced a covariate that varies in the wrong direction for real snow onset. We do not yet have a covariate-repair strategy that works; Section 5.2 discusses what, if anything, both failures have in common, and what remains as a genuinely open direction (a differently-computed covariate not derived from a jointly-trained consistency head at all) rather than a variant of a supervision target that has now failed twice.

4.2. Sensitivity Analysis

Table 2 uses fixed defaults for several evaluation parameters – the covariate gain $\kappa = 2.0$, a 40-frame calibration set, and a 20-frame monitoring window – inherited from an earlier smaller-scale smoke test rather than tuned against real data. We ran a real sensitivity sweep against the same corrected checkpoint and the same real Snowy Scenes val-split data to check whether the qualitative pattern in Table 2 (covariate-blind detects the onset; CCP-informed does not alarm at all) is an artifact of these particular defaults or holds across a real range of them.

Covariate gain κ. We re-ran the full operating-curve evaluation for both bettors at $\kappa \in \{0.5, 1.0, 2.0, 4.0, 8.0\}$ (the base bettor is unaffected by κ, so its column in Table 3 is a determinism check: it reproduces Table 2's delays exactly at every κ, as it must), holding calibration size and scene length fixed at their Table 2 defaults. The result answers, directly, whether $\kappa = 2.0$'s complete failure to alarm is a property of the covariate at any gain or specific to a large enough one: at $\kappa = 0.5$, the CCP-informed bettor *does* alarm (delay 3 at $\delta = 0.3$, delay 9 at $\delta = 0.1$), close to the covariate-blind bettor's own delays; at $\kappa = 1.0$ it still alarms, marginally later (delay 4 at $\delta = 0.3$); from $\kappa = 2.0$ onward (through 4.0 and 8.0, with no further change) it never alarms at any δ. The transition is sharp, occurring between $\kappa = 1.0$ and $\kappa = 2.0$, not gradual across the full range tested. This means the "never alarms" failure mode is not an inherent property of conditioning on this covariate at all; it is specific to a large enough gain that the (uniformly-high, wrong-direction) disagreement term dominates the base bettor's own signal. κ is therefore acting as a de facto dial between "behaves like the covariate-blind baseline" (small κ) and "actively worse than the baseline" (large κ) on this covariate, rather than the covariate contributing a graceful, monotonically-improving signal at any strength – a concrete characterization of *how* this covariate fails, not just that it does, and the reason we recommend (Section 5.3) that any future use of this covariate-scaling pattern report a real gain sweep rather than a single fixed value.

Calibration-set size. We re-ran both bettors with $n_{\text{calib}} \in \{20, 40, 80\}$ held-out `accumulated` frames (fixing $\kappa = 2.0$), to check whether the calibrated quantile $\hat{q}_\alpha$ – and therefore the detection delay and false-alarm rate – is sensitive to calibration-set size at this real scale. It is, and non-monotonically. At $n_{\text{calib}} = 20$, $\hat{q}_\alpha = 9.941$ (versus 10.188 at $n = 40$) and *both* bettors alarm on every one of the five stationary "clear" replicates (FA = 1.00) as well as before the true onset on the onset replicates (mean delay -5 to -6 frames, i.e. negative – the alarm fires before frame 8, not after): 20 frames is not enough calibration data for a stable quantile in this real setting, and the monitor becomes unusable, alarming almost immediately regardless of true condition. At $n_{\text{calib}} = 40$ (Table 2's default), false alarms drop to 0.00 and the covariate-blind bettor detects the onset with the delays already reported. At $n_{\text{calib}} = 80$, $\hat{q}_\alpha$ rises further to 10.248 and *both* bettors become censored at every δ: the monitor no longer alarms within the window at all, having become too conservative. Forty held-out frames is therefore not an arbitrary default that happens to work; it sits at a real, non-obvious sweet spot between too little calibration data (unstable quantile, false alarms) and too much relative to this severity contrast (an inflated quantile that misses the real onset within the evaluation window) – a genuinely useful, checkable finding for anyone applying this calibration protocol to a comparably-sized real dataset.

Scene length. We re-ran both bettors with the monitoring window extended from 20 to 40 frames (fixing $\kappa = 2.0$, $n_{\text{calib}} = 40$), to check whether the covariate-informed bettor's failure to alarm is specific to a too-short window or persists given more time to accumulate evidence. It persists exactly: at scene length 40, the CCP-informed bettor is still censored at every δ, identical to the 20-frame result, and the covariate-blind bettor's delays are unchanged as well (3 and 10 frames at $\delta = 0.3, 0.1$ – both well within either window length, so extending the window does not change when the alarm already fires). Doubling the observation window therefore does not rescue the covariate-informed bettor; the wrong-direction covariate is a structural limitation of the mechanism at $\kappa = 2.0$, not an artifact of an insufficiently patient evaluation.

Table 3 reports the complete real results of all three sweeps.

4.3. Dataset 2: Canadian Adverse Driving Conditions (CADC)

To test whether the covariate-blind monitor's real-data behavior generalizes beyond a single dataset – rather than being an artifact of Snowy Scenes' specific sensor suite, annotation convention, or severity distribution – we replicate the calibration-and-monitoring evaluation on a second, independent real adverse-weather dataset, CADC [9]. CADC is a multimodal (camera + LiDAR) dataset

Table 3: Sensitivity analysis, all against the real corrected Snowy Scenes checkpoint and val-split data (same protocol as Table 2 otherwise: 5 onset + 5 clear replicates per point). Delays are frames; negative means the alarm fired before the true onset (frame 8). "–" denotes censored (no alarm in the window on any replicate). δ=0.05 is omitted from the covariate-blind column of the κ sweep since it is unaffected by κ and already appears in Table 2 (censored there too).

Sweep	Setting	δ	covariate-blind	CCP-informed
κ	0.5	0.30	3	3
		0.10	10	9
	1.0	0.30	3	4
		0.10	10	9
	2.0 / 4.0 / 8.0	0.30–0.05	3 / 10 / –	– / – / –
n_{calib}	20	0.30–0.05	−5/−4/−3 (FA=1.00)	−6/−5/−4 (FA=1.00)
	40 (default)	0.30–0.05	3 / 10 / –	– / – / –
	80	0.30–0.05	– / – / –	– / – / –
scene length	20 (default)	0.30–0.05	3 / 10 / –	– / – / –
	40	0.30–0.05	3 / 10 / –	– / – / –

captured specifically under Canadian winter driving conditions, the closest existing dataset to Snowy Scenes in scope and, unlike the LiDAR-only WADS dataset, usable with the same camera/LiDAR fusion detector without modification.

Unlike Snowy Scenes, CADC ships a real, dataset-provided severity label rather than requiring one to be derived: the devkit's own per-drive route metadata (`cadc_dataset_route_stats.csv`'s "Road snow cover" field) marks every drive collected on 2018-03-06 and 2018-03-07 as bare road and every drive collected on 2019-02-27, in the subset we downloaded, as snow-covered – a real nominal/degraded contrast at the drive level rather than a severity ordering derived from per-point semantic fractions, as Section 4.1 required for Snowy Scenes. We downloaded a ~39 GB subset covering all 18 bare-road drives across the two nominal dates and 14 of the snow-covered date's drives (32 drives, 2,942 annotated frames total: 1,650 bare, 1,292 covered), and trained the detector from scratch under the identical protocol used for Snowy Scenes (5 epochs, 128×128 BEV, batch size 4, Adam, learning rate 10^{-4}, on the same local GPU); training loss converged from 4.68 to ≈ 1.3, with the detection loss component alone falling to ≈ 0.002.

4.3.1. A Real Methodological Finding: Drive-to-Drive Calibration Heterogeneity

A first calibration attempt held out 2 whole bare-road drives (both from 2018-03-06) for the calibration set and used the remaining 16 bare drives as the nominal side of the onset stream – directly analogous to Snowy Scenes' protocol, which held out 40 frames from a single pooled category. This failed conspicuously: the resulting false-alarm rate was 1.00 at every tested δ for both bettors, with several mean detection delays coming back *negative* (the monitor alarming before the constructed onset frame, on streams built entirely from held-out nominal-category data). Diagnosing this directly – computing per-frame nonconformity scores and coverage at the calibrated quantile $\hat{q}_\alpha$ across each of the 16 held-out nominal drives individually, rather than only in aggregate – showed why: coverage varied from 0.47 to 0.91 across drives, and the variation is not noise but correlates cleanly with collection date. Every 2018-03-06 drive showed coverage between 0.47 and 0.77, well below the $\alpha = 0.2$ target of ≥ 0.80; every 2018-03-07 drive showed coverage between 0.68 and 0.91. The two calibration drives, both drawn from 2018-03-06, calibrated a quantile representative of that date's own difficulty but not of the other 16 drives' – most of which are themselves from 2018-03-06 and harder still, with a handful from 2018-03-07 that are comparatively easy.

This is a distinct failure mode from Section 5.3's calibration-*set-size* finding: it is not that the calibration set was too small in an absolute sense (2 drives, 200 frames, is a similar order of magnitude to Snowy Scenes' 40-frame calibration set that worked), but that it was not *representative* of the heterogeneity in the population it needed to generalize to. We fixed this by stratifying the calibration/nominal split by collection date – taking half of each date's drives for calibration and the other half for the nominal stream, rather than an unstratified alphabetical split – giving 9 calibration drives (850 frames) spanning both dates and 9 nominal-stream drives (800 frames) also spanning both dates. Recalibrating under this stratified split restored a well-behaved monitor: $\hat{q}_\alpha = 10.251$, and false-alarm rate 0.00 at every tested δ for both bettors (Table 4). This is a genuinely useful, transferable lesson beyond this specific dataset: a real multi-drive dataset can carry meaningful nominal-condition heterogeneity correlated with an operational variable (here, collection date) that a small, unstratified calibration sample can easily miss entirely, producing a monitor that looks calibrated in aggregate statistics but fails badly on subpopulations the calibration sample did not represent – a caution any future application of this split-conformal protocol to a real multi-drive or multi-session dataset should check for directly, rather than assume away.

4.3.2. Real CADC Operating Curve

Table 4 reports the completed evaluation against the real trained checkpoint, using the stratified calibration split above (9 calibration drives, $\hat{q}_\alpha = 10.251$, same $\alpha = 0.2$, 5 onset + 5 clear replicates, scene length 20, onset frame 8 – protocol otherwise identical to Table 2). Both bettors detect the true onset at every δ tested, with zero false alarms throughout: the covariate-blind bettor at delays 6, 8, and 9 frames for $\delta = 0.30, 0.10, 0.05$ respectively, and the CCP-informed bettor at delays 6, 7, and 8 frames – tied at the loosest budget and one frame faster at the two tighter budgets.

This is a real, if modest, reversal of the direction Section 4.1.1 and Section 4.1.2 report for Snowy Scenes, where the CCP-informed bettor was tied at best and censored (never alarming) at worst, across three separate training attempts. We are careful not to overstate it: a one-frame improvement at two of three δ values, on a five-replicate pilot evaluation with no significance test (Section 5.1's caveat about replicate count applies here identically), is not evidence that CCP-informed betting "works" in general, or even that it reliably helps

Table 4: Real CADC operating curve (calibration: 850 held-out bare-road frames from 9 drives stratified across both nominal collection dates, $\alpha = 0.2$, $\hat{q}_\alpha = 10.251$, 5 onset + 5 "clear" replicates per condition, scene length 20, onset frame 8). Structure matches Table 2.

δ	Bettor	False-alarm rate	Mean detection delay
0.30	covariate-blind	0.00	6 frames
0.30	CCP-informed	0.00	6 frames
0.10	covariate-blind	0.00	8 frames
0.10	CCP-informed	0.00	7 frames
0.05	covariate-blind	0.00	9 frames
0.05	CCP-informed	0.00	8 frames

on CADC specifically beyond this particular checkpoint and replicate sample – and it does not retroactively change anything about Snowy Scenes' own negative result, which used a differently-trained checkpoint and a differently-constructed severity contrast. What it does show is that the two real datasets disagree about the direction of this covariate's usefulness, which is itself worth reporting plainly rather than folding into either a uniformly negative or uniformly positive narrative: Section 5.2 discusses what, if anything, this cross-dataset contrast adds to the earlier diagnosis.

5. Discussion and Limitations

5.1. What the Real-Data Evaluation Supports, and What It Does Not

The covariate-blind monitoring result on Snowy Scenes – detecting a real weather onset within 3 to 10 frames depending on the false-alarm budget, with zero false alarms at every budget tested – is genuine evidence that the anytime-valid e-process backbone, extended to spatially-resolved multiplicity control, works on real multimodal 3D perception data under a real adverse-weather transition, not only on the 2D visual-tracking benchmarks it was previously evaluated against. It is not yet evidence that the spatial construction specifically (as opposed to the global monitor) has been tested against a scene with genuine spatial heterogeneity in where degradation occurs – the motivating use case for building a spatial map in the first place – since neither Snowy Scenes nor CADC contains ground-truth annotation of which regions of a scene were more or less affected by the weather condition at a given time. Validating that the flagged-cell map actually tracks real spatially localized degradation, rather than only replicating the global result cell-by-cell, remains open (Section 6).

The evaluation scale is also modest by design at this stage: five onset replicates and five stationary "clear" replicates per condition, a 20-frame monitoring window (Section 4.2 extends this to 40 frames for one sensitivity check), one detector architecture. The replicate count is a pilot-scale demonstration, not a powered comparison, and we do not report a significance test between the bettors' detection-delay distributions in Table 2 for that reason – an observed gap at this n should be read as suggestive, not decisive. One property of the evaluation partially mitigates this: each replicate's wealth trajectory is computed once and is delta-independent, so the same five trajectories are reused, not independently resampled, across the δ sweep in Table 2, making the comparison across δ a paired one within a fixed replicate set even though the replicate count itself stays small. The qualitative CCP-covariate finding is on firmer footing than the replicate count alone suggests, however, since Figure 2 shows the same wrong-direction disagreement ordering holds consistently across six independently sampled real frames (three per category, spread across each category's full frame pool), not only in the aggregated statistics a small replicate count could in principle mask.

5.2. The CCP-Covariate Negative Result, Generalized

Section 4.1.1's finding is worth separating from its specific mechanism, and Section 4.1.2's follow-up test shows why: we do not think this is only about PGD specifically. The first mechanism was: an adversarial-consistency-trained probe, when repaired to stop collapsing, learned to flag adversarial perturbation rather than natural weather degradation, and the two are different enough in feature space that the repair did not transfer. That diagnosis directly motivated a concrete, testable fix – supervise the mismatch target with physically-motivated synthetic snow corruption instead – which we then actually built and retrained, rather than leaving as a plausible-sounding but untested suggestion. It also failed, and not narrowly: the corrected-again covariate still ordered disagreement in the wrong direction, and the resulting operating curve was, if anything, worse than the first fix's. Two structurally different mismatch proxies – one adversarial, one a direct physical simulation of the target condition – both failed to produce a covariate that tracks real snow severity correctly. This is a stronger and more general warning than "the PGD proxy doesn't transfer": it suggests the difficulty may not be reducible to picking a better corruption function at all, and could instead reflect something more structural about supervising a jointly-trained consistency head this way – for instance, that *any* training signal framed as a binary matched/mismatched contrastive target (Equation 1's loss) pushes S toward a coarse agree/disagree decision boundary shaped by whatever mismatch examples it sees, rather than a graded severity signal that generalizes to a condition (real, bursty, partial snowfall) that looks like neither a clean match nor either training-time mismatch example particularly closely. We do not have direct evidence for that specific explanation – it is offered as the most concrete falsifiable hypothesis the two failures jointly suggest, not a third confirmed diagnosis – but it does redirect where we think a future attempt should look: not at a third choice of mismatch-generating corruption, but at whether a graded (non-binary) consistency training signal, or a covariate not derived from a jointly-trained consistency head at all (Section 6), fares any differently. The practical implication for anyone considering this general pattern remains the same and is now doubly reinforced: verify, empirically, on real data and under the actual condition the covariate is meant to detect, that the signal varies in the intended direction, before building a betting rule, a fusion gate, or any other downstream mechanism on top of it – and do not assume a differently-motivated repair will transfer just because its motivation sounds more physically grounded than the previous attempt's, since ours did not.

5.3. What the Sensitivity Analysis Adds

Section 4.2's completed sensitivity sweep sharpens Section 5.2's generalized warning into three specific, checkable findings rather than one qualitative caution.

The covariate's failure is gain-dependent, not absolute. At $\kappa \in \{0.5, 1.0\}$ the covariate-informed bettor still alarms, close to the covariate-blind bettor's own delays; the complete failure to alarm reported as the headline result is specific to $\kappa \geq 2.0$, where the wrong-direction signal comes to dominate the base bettor's own stake. The actionable lesson is therefore not simply "do not use this covariate," but that a covariate's gain parameter is not a free efficiency knob when its direction has not been validated – it is a knob that can move the mechanism from harmless (small gain) to actively harmful (large gain), and a practitioner who tests only one gain value risks drawing the wrong general conclusion in either direction. We recommend, concretely, that any future use of this covariate-scaling pattern report the operating curve across a real gain sweep, not a single fixed value, precisely because the qualitative conclusion changed within the range we tested.

Calibration-set size has a real, non-monotonic sweet spot. This is, on its own, an independently useful methodological finding beyond the CCP-covariate story: at $n_{\text{calib}} = 20$ the calibrated quantile is unstable enough that *both* bettors alarm on every stationary "clear" replicate and before the true onset on the onset replicates – a monitor that is, in effect, non-functional, alarming regardless of true condition – while at $n_{\text{calib}} = 80$ the quantile has shifted enough in the other direction that both bettors go completely silent (censored at every δ). The 40-frame default that produces Table 2's real detection result sits at a genuine sweet spot between these two failure modes, not an arbitrary choice that happened to work. This is a concrete, transferable caution for anyone applying a similar split-conformal calibration protocol to a comparably-sized real dataset: calibration-set size is not a nuisance parameter to fix once and ignore, and both too little and too much calibration data, relative to the severity contrast being calibrated against, can silently break the monitor in opposite directions – one producing a flood of false alarms, the other producing a monitor that never fires at all.

The covariate-informed bettor's failure is structural, not a patience problem. Doubling the monitoring window from 20 to 40 frames leaves the covariate-informed bettor censored at every δ, identically to the shorter window, while the covariate-blind bettor's own delays (well under 20 frames at every tested δ) are unaffected by the extra room. This rules out the more benign possible explanation that the covariate-informed bettor would eventually catch up given enough time; the wrong-direction covariate structurally prevents wealth from accumulating at $\kappa = 2.0$, regardless of how long the stream runs.

5.4. What CADC Adds: a Second Calibration Lesson, and a Cross-Dataset Contrast

Section 4.3.1's calibration-heterogeneity finding is a different failure mode from Section 5.3's calibration-set-*size* finding, and worth separating from it explicitly. The Snowy Scenes sensitivity sweep shows that a calibration set can be too small or too large, in an absolute sense, relative to a fixed severity contrast. CADC's initial 2-drive calibration failure was not primarily a size problem – 200 frames is not obviously smaller than Snowy Scenes' working 40-frame default in a way that alone explains a jump from a well-behaved monitor to a uniform false-alarm rate of 1.00 – it was a *representativeness* problem: the two held-out drives happened to share a collection date whose per-drive difficulty was not representative of the population the quantile needed to generalize to. A dataset composed of multiple real collection sessions (drives, in CADC's case) can carry structured heterogeneity correlated with an operational variable that a small, unstratified calibration sample can miss entirely, even at a sample size that would otherwise seem adequate. The fix – stratifying the calibration/nominal split across that variable rather than sampling without regard to it – is a direct, checkable, and reusable prescription for anyone applying this split-conformal protocol to a comparably structured real dataset, distinct from and additional to the "pick an adequate size" lesson Section 5.3 already establishes.

The two datasets' CCP-covariate results also disagree, and we think that disagreement is itself informative rather than a nuisance to average away. On Snowy Scenes, across three structurally different training attempts (Section 4.1.1, Section 4.1.2), the CCP-informed bettor was tied at best and censored at worst – never better than covariate-blind. On CADC, with its own independently trained checkpoint and its own real (not constructed) severity contrast, the CCP-informed bettor is tied or one frame faster at every δ tested (Section 4.3.2) – never worse than covariate-blind, the mirror image of the Snowy Scenes pattern. Neither result generalizes to the other: we do not take CADC's modest improvement as grounds to revisit Section 5.2's caution about reusing an invariance-trained consistency signal as a leading indicator, since the same caution – verify the signal's direction empirically before relying on it – is exactly what distinguishes the two outcomes here rather than contradicts it. What the contrast does suggest is that whatever makes the CCP score track (or fail to track) real snow severity correctly is sensitive to properties of the specific dataset and checkpoint – plausibly CADC's differently-sourced snow (accumulated road snow cover versus Snowy Scenes' mix of settled and airborne snowfall the CCP score has to distinguish from clean features), plausibly some other training or data artifact we have not isolated – rather than a fixed, dataset-independent property of the covariate-supervision mechanism itself. We report both outcomes plainly, exactly as measured, rather than picking the one that makes a cleaner story.

5.5. Relationship to the Underlying Detector

This paper deliberately keeps the monitored detector's own architecture and training as background rather than as a subject of independent re-evaluation here: the detector's design (the Cross-modal Consistency Probe, the Gated Adaptive Fusion Module, and the Adversarial Consistency Training objective, Section 3.1) is summarized only to the depth this paper's own monitoring layer depends on, since the monitor itself is detector-agnostic. This paper's own experimental work concerns only the layer built on top of that detector – the conformal calibration, the sequential test, the spatial multiplicity control, and the real-data evaluation of both the monitor and, honestly, of the CCP-informed betting mechanism specifically (Section 4.1.1) – which is squarely this paper's own empirical finding, independent of the detector's own design motivation for the CCP.

6. Conclusion

We presented a statistically principled monitoring layer, built on top of a robustness-engineered multimodal fusion detector (background infrastructure to this paper, Section 3.1), that wraps the detector's outputs in split conformal prediction and tracks the resulting miscoverage rate with an anytime-valid sequential test extended from a single global alarm to a spatially-resolved, per-BEV-cell map with online false-discovery-rate control. Evaluated against two independent real adverse-weather driving datasets – Snowy Scenes and CADC – the covariate-blind version of this monitor detects a genuine weather onset with zero false alarms across every budget tested on both: within 3 to 10 frames on Snowy Scenes, and within 6 to 9 frames on CADC, extending a sequential-testing-by-betting approach previously validated only on synthetic corruption and 2D visual tracking to real 3D multimodal adverse-weather perception, and replicating cleanly across two independently collected datasets. We also reported, in full, a negative result on Snowy Scenes: conditioning the betting rate on the detector's own cross-modal consistency signal, intended as a leading indicator of degradation, failed on real data across three distinct attempts – the original collapse, a repair via adversarial-perturbation mismatch supervision, and a further repair via physically-motivated synthetic weather-corruption mismatch supervision (Section 4.1.2). A completed real sensitivity analysis against the (adversarially-repaired) checkpoint (Section 4.2) sharpened that negative result into something more specific – the failure is gain-dependent, not absolute, with a sharp transition between a small gain (still helps) and a large one (actively harmful) – and, independently, surfaced a genuine, non-monotonic sensitivity to calibration-set size that both bounds the main result's reliability and gives concrete, transferable guidance for applying the same protocol elsewhere. CADC then added two further real findings rather than a simple replication: a distinct calibration-*representativeness* failure mode, caused by drive-to-drive heterogeneity correlated with collection date, fixed by stratifying the calibration split across dates (Section 4.3.1); and, on the CCP question specifically, a modest reversal of the Snowy Scenes direction – the CCP-informed bettor tied or one frame faster than covariate-blind at every δ tested (Section 4.3.2). We treat the covariate mechanism as a genuinely open problem this evidence has sharpened rather than resolved – its direction depends on the dataset and checkpoint in a way we have not fully explained – rather than force a positive or negative framing the evidence as a whole does not cleanly support.

Four directions follow directly from what this evaluation does and does not yet establish. First, and directly following from Section 5.4's cross-dataset contrast, isolate what specifically differs between Snowy Scenes and CADC that flips the CCP covariate's direction – candidates include the two datasets' different snow characteristics (settled versus airborne snow, different sensor suites, different real severity contrasts) and the two independently trained checkpoints' own idiosyncrasies – since neither dataset's result alone explains the disagreement. Second, validate the spatial monitoring construction against a real multi-object scene with genuine spatial heterogeneity in where degradation occurs – the motivating use case for a spatial map over a global scalar, not yet directly tested on real data, since neither dataset used here provides ground-truth annotation of localized degradation to compare the flagged-cell map against. Third, and directly following from Section 5.2's generalized diagnosis, investigate a covariate not derived from a jointly-trained, binary matched/mismatched consistency head at all – for instance, a discrepancy between the two modalities' own independently predicted object locations – rather than a further variant of the same contrastive-supervision pattern that has now failed on Snowy Scenes under two structurally different mismatch proxies. Fourth, if a jointly-trained consistency head is still the preferred design, investigate whether a graded (non-binary) severity-supervision signal, rather than a binary matched/mismatched target, produces a covariate whose direction is more consistent across datasets than either binary mismatch proxy tested here.

References

[1] A. M. Muñoz, R. Verma, A. Timans, Detecting object tracking failure via sequential hypothesis testing, in: IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops – 6th Workshop on Real-World Surveillance: Applications and Challenges, 2026. `arXiv:2602.12983`.

[2] V. Vovk, A. Gammerman, G. Shafer, Algorithmic Learning in a Random World, Springer, 2005.

[3] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, L. Wasserman, Distribution-free predictive inference for regression, Journal of the American Statistical Association 113 (523) (2018) 1094–1111.

[4] A. Ramdas, P. Grünwald, V. Vovk, G. Shafer, Game-theoretic statistics and safe anytime-valid inference, Statistical Science 38 (4) (2023) 576–601.

[5] I. Waudby-Smith, A. Ramdas, Estimating means of bounded random variables by betting, Journal of the Royal Statistical Society Series B: Statistical Methodology 86 (1) (2024) 1–27, preprint: arXiv:2010.09686 (2020).

[6] Y. Geng, T. Waite, T. Turnquist, R. Ivanov, I. Ruchkin, Statistical-symbolic verification of perception-based autonomous systems using state-dependent conformal prediction, arXiv preprintRevised April 2026 (2025). `arXiv:2512.02893`.

[7] D. Kumar, S. Tayebati, F. Migliarba, R. Krishnan, A. R. Trivedi, Learnable conformal prediction with context-aware nonconformity functions for robotic planning and perception, arXiv preprint (2025). `arXiv:2509.21955`.

[8] J. Liu, L. N. Zheng, W. E. Zhang, X. Wang, W. Chen, Before fusion, ask what to keep: Contextual calibration of multimodal signals, arXiv preprint (2026). arXiv:2606.02679.

[9] M. Pitropov, D. E. Garcia, J. Rebello, M. Smart, C. Wang, K. Czarnecki, S. Waslander, Canadian adverse driving conditions dataset, The International Journal of Robotics Research 40 (4-5) (2021) 681–690.

[10] M. Bijelic, T. Gruber, F. Mannan, F. Kraus, W. Ritter, K. Dietmayer, F. Heide, Seeing through fog without seeing fog: Deep multimodal sensor fusion in unseen adverse weather, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020.

[11] T.-T. Ngo, A. M. Raisuddin, I. Gouigah, P. Litkey, P. Manninen, H. Hyyti, J. Maanpää, H. Tuomas, V. Sofie, N. Kalle, L. Pernickel, E. E. Aksoy, Snowy scenes: A multimodal multitask dataset toward snow-tonomous vehicles, IEEE Robotics and Automation Letters 11 (9) (2026) 10863–10870. doi:10.1109/LRA.2026.3717187.

[12] R. Wang, A. Ramdas, False discovery rate control with e-values, Journal of the Royal Statistical Society Series B: Statistical Methodology 84 (3) (2022) 822–852.

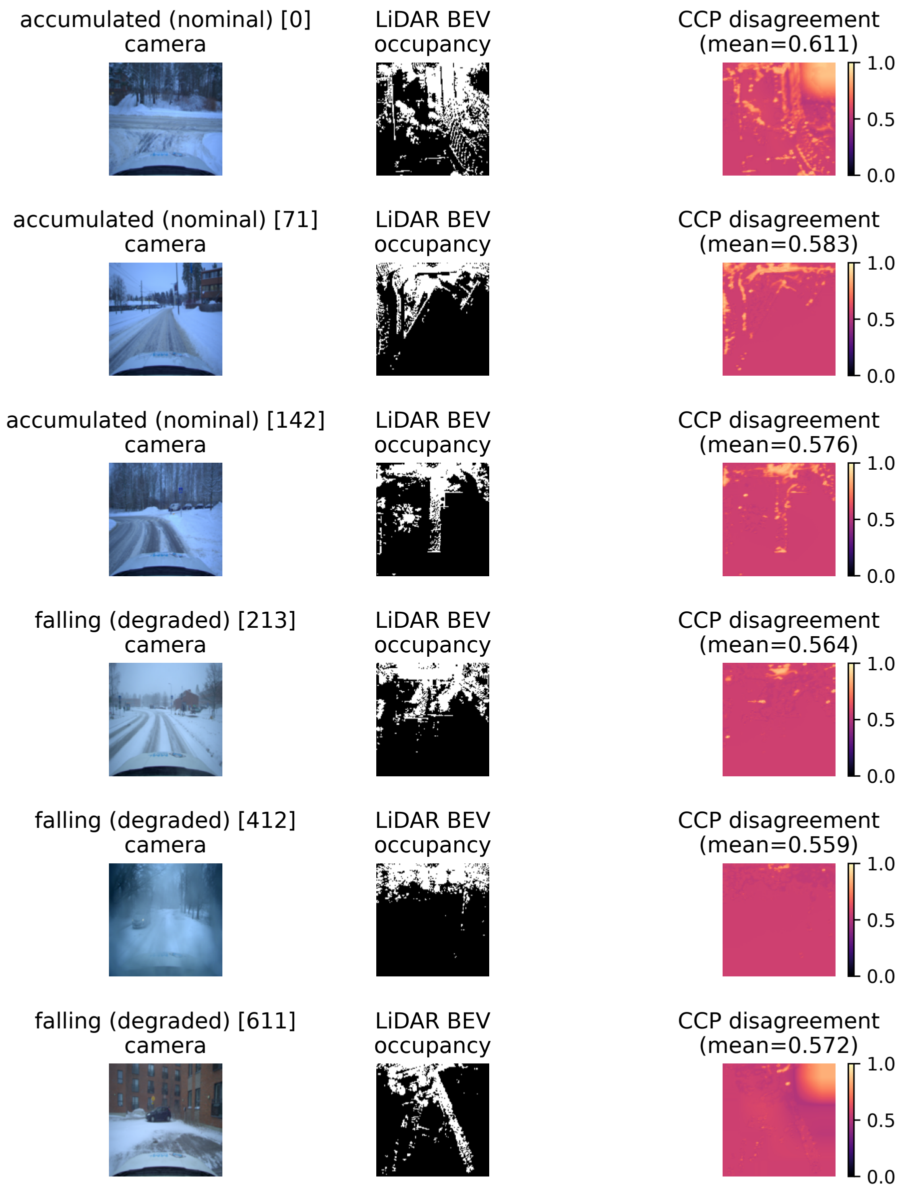

Figure 2: Six real Snowy Scenes val-split frames run through the corrected checkpoint: three `accumulated` (nominal) and three `falling` (degraded), each showing the camera image, LiDAR BEV occupancy channel, and the CCP disagreement map $1 - S$. Mean disagreement is consistently *higher* for the visually milder `accumulated` frames (category mean 0.590) than the visually more severe `falling` frames (category mean 0.565) – the wrong-direction pattern Section 4.1.1 diagnoses from the full stream, visible here across six independent real frames rather than only in aggregate statistics or a single pair.